

NMJ21803

TUTORIAL CHAPTER 3

1. Consider a semi-infinite filamentary conductor carrying a current of 5 A in the positive-y direction. Find the magnetic flux density, $\mathbf{B}$, at P (4, 6, 0) using Biot-Savart's law.
2. The conducting triangular loop carries a current of 10 A. Find $\mathbf{H}$ at (0, 0, 5) due to side 1 of the loop.
3. A circular loop located on $x^2 + y^2 = 9$, $z = 0$ carries direct current of 10 A along $\hat{\phi}$. Determine $\mathbf{H}$ at (0, 0, 4) and (0, 0, -4).
4. Planes $z = 0$ and $z = 4$ carry current $\mathbf{K} = -\hat{x}10 \text{ A/m}$ and $\mathbf{K} = \hat{x}10 \text{ A/m}$. Determine $\mathbf{H}$ at (1, 1, 1) and (0, -3, 10).
5. A toroid of circular cross section whose center is at the origin and axis the same as z-axis has 1000 turns with $\rho_o = 10 \text{ cm}$, $a = 1 \text{ cm}$. If the toroid carries a 100 mA current, find $\mathbf{H}$ at (3 cm, -4 cm, 0) and (6 cm, 9 cm, 0).
6. In free space, the magnetic flux density is given by:

$$\mathbf{B} = \hat{x}2x^2y^2z - \hat{y}2xy^3z + \hat{z}xy^2z^2 \text{ (Wb/m}^2\text{)}$$

- (a) Prove that $\mathbf{B}$ is a magnetic field.
- (b) Evaluate the flux, Φ , through a conductor's surface of $x = 1$, $0 \leq y \leq 2$, $0 \leq z \leq 4$.
- (c) Determine the current density, $\mathbf{J}$, crossing through the surface of (b).

All distances are given in meters.

7. A magnetic vector potential is given by:

$$\mathbf{A} = \hat{\mathbf{r}} \frac{r^3}{3} \cos 2\phi \text{ (Wb/m)}$$

Determine the total magnetic flux, Φ , crossing the surface of $2 \leq r \leq 4$, $\pi/3 \leq \phi \leq \pi/2$, $z = 3$

All distances are in meters.

8. The semicircular conductor shown carries a current I . The closed circuit is exposed to a uniform magnetic field, $\mathbf{B} = \hat{\mathbf{y}}B_0$. Determine (a) the magnetic force $\mathbf{F}_1$ on the straight section of the wire and (b) the force $\mathbf{F}_2$ on the curved section.

9. A magnetic field is defined by:

$$\mathbf{H} = \hat{\phi} 5r^2 \cos \phi \text{ (A/m)}$$

It is exposed in free space over an infinite straight wire located at $r = 3\text{m}$ and $\phi = 45^\circ$. The wire is conducting a current of 2.5 A.

- (a) Determine the magnetic force acting on the wire considering $-5\text{m} \leq z \leq 5\text{m}$.
(b) Prove that the magnetic force does no work when the wire is turned once around the z-axis in clockwise direction.
10. The solenoid with 100 turns carries a current of 5 A. Using soft iron as core with $\mu_r = 5000$.
- (a) Determine the self-inductance of the solenoid.
(b) Calculate the magnetic energy stored in the solenoid.